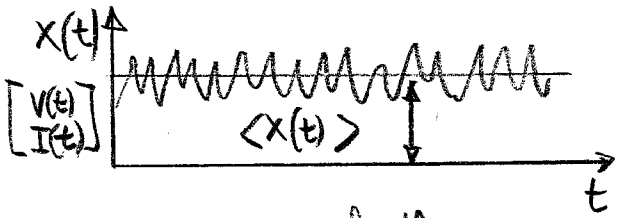
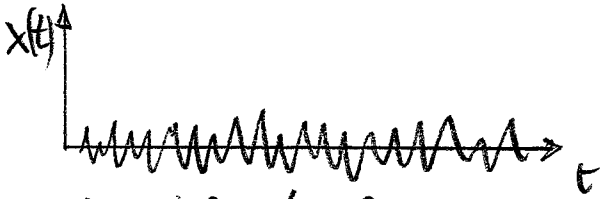
NOISE — Stationary Process — Statistical properties of Noise (1)



Remove $\langle x(t)\rangle$ →

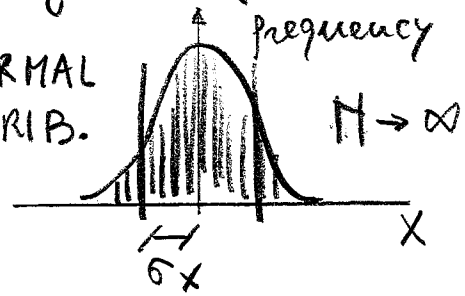


x_i samples

$$\langle x\rangle = \frac{1}{M}\sum x_i = 0$$

Istogram of the samples: can be described by a Probability density function

NORMAL DISTRIB.



$$\rightarrow \quad PDF(x) = \frac{\exp\left[-\frac{1}{2}\frac{x^2}{\sigma_x^2}\right]}{\sqrt{2\pi}\,\sigma_x} \qquad \int_{-\infty}^{+\infty} PDF(x)\,dx = 1$$

$PDF(\bar{x})\,dx$ = probability of finding a sample from $\bar{x} \rightarrow \bar{x}+dx$

$$\sigma_x = \sqrt{\frac{1}{M}\sum(x_i - \cancel{\langle x\rangle})^2} = \sqrt{\frac{1}{M}\sum x_i^2} = \sqrt{\int_{-\infty}^{+\infty} x^2\, PDF(x)\,dx}$$

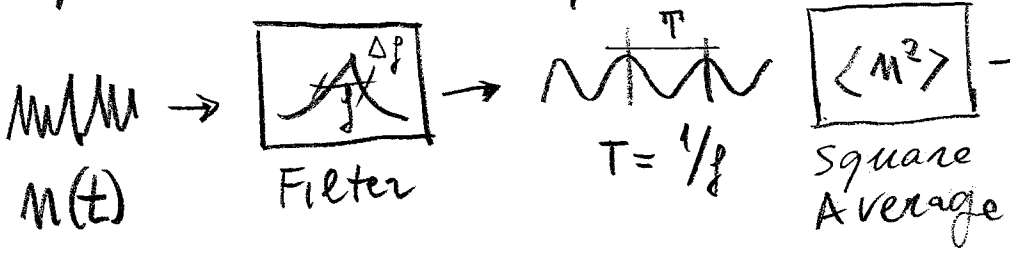
STD — 68% of samples $-\sigma_x, +\sigma_x$

$$\sigma_x^2 = \frac{1}{M}\sum x_i^2$$ Variance $\sigma_x^2 = \langle x^2\rangle$ mean square value of X

If X is Voltage or current $\sigma_x^2 \propto$ Average Power of noise $\sigma_x^2 = \langle P_x\rangle$

Spectral properties of Noise $X \rightarrow M$

Spectrum analyzer



$$\rightarrow \langle M^2(t)\rangle\Big|_{f\to f+\Delta f} \rightarrow \boxed{S(f) = \frac{\langle M(t)^2\rangle}{\Delta f}\Big|_{f\to\Delta f}}$$

$\boxed{\text{Noise Power Spectrum}}$

↓ Noise Power spectral density

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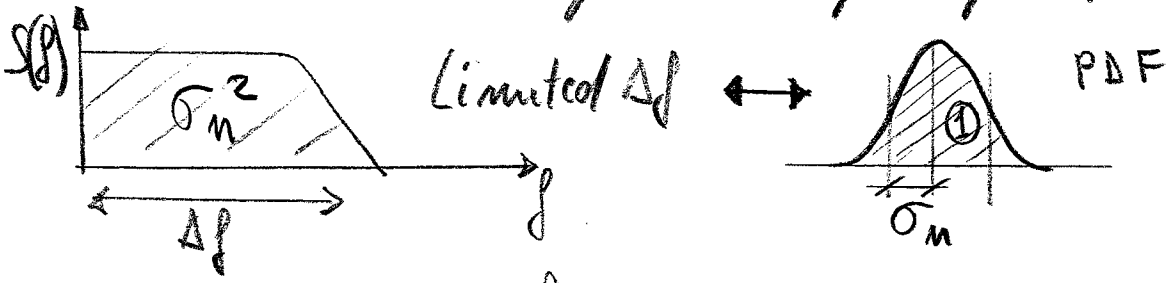
$$\int_0^{\infty} S(f)\,df = \langle n^2(t)\rangle = \langle P_n\rangle = \sigma_n^2$$

$$\langle [A\sin(2\pi f_1 t) + B\sin(2\pi f_2 t)]^2\rangle = \langle A^2\sin^2(2\pi f_1 t)\rangle + \langle B^2\sin^2(2\pi f_2 t)\rangle = \frac{A^2}{2} + \frac{B^2}{2}$$

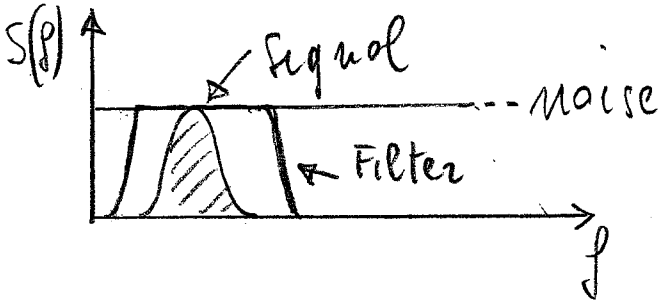
2

The integral of the noise power spectrum gives the mean square value of the noise, that is the noise average power [Parseval Theorem]

The equality $\int_0^{+\infty} S(f)\,df = \sigma_n^2$ establishes a relationship between statistical and spectral properties of noise [Variance $\leftrightarrow$ Power]



If the bandwidth of the signal is smaller than the bandwidth of the noise we can improve S/N by using an appropriate spectral filter tuned to the bandwidth of the signal



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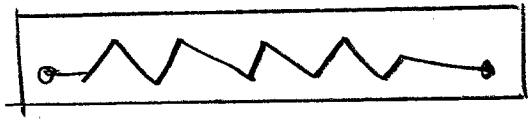
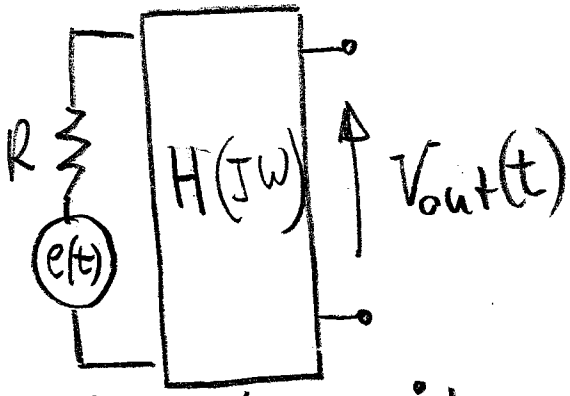
Model for Thermal noise



R

A real resistor is modelled by the series of an ideal resistor and a noise electromotive force generator $e(t)$



equivalent circuit

The noise of the resistor is transferred to the output by a linear network with Transfer function $H(j\omega)$

The mean square value of the noise in the frequency interval $f \to f+df$ is:

$$\langle e^2(t)\rangle\Big|_{f\to f+df} = S(f)\,df$$ [How is this quantity transferred to the output?]

In general: $$\langle Out^2(t)\rangle\Big|_{f+df} = \langle in^2(t)\rangle_{f+df} \cdot |H(j\omega)|^2 \circledast \quad [\omega = 2\pi f]$$

Let's assume that the linear network is made by a capacitor $\dashv\vdash C$

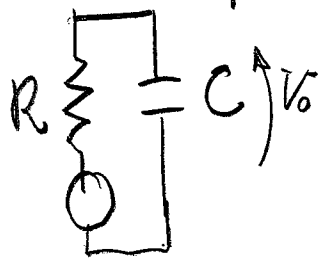
$$H(j\omega) \longrightarrow \frac{1}{1+j\omega RC}$$ [Dividing ⊛ by df]

$$S_{out}(f) = S_{in}(f)\cdot|H(j\omega)|^2 = S_{in}(f)\cdot\frac{1}{1+\omega^2R^2C^2}$$

$$Z = R + \frac{1}{j\omega C} = \frac{1+j\omega RC}{j\omega C}$$

$$I = \frac{e}{Z} = \frac{e\,j\omega C}{1+j\omega RC}$$

$$V_0 = I\cdot Z_C = \frac{e}{1+j\omega RC}$$



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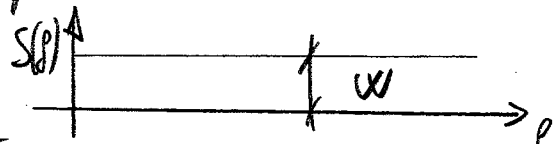
Let's assume a uniform power spectrum for the noise electromotive force

$S_{in}(f) = W$ constant $\forall f$ → White noise



<u>Mean square value of the noise at the output</u> = Average noise power $\langle P_n \rangle$

$$\langle V_{out}^2(t) \rangle = \int_0^\infty S_{out}(f)\, df = \int_0^\infty S_{in}(f) \cdot |H j2\pi f|^2 df = W \int_0^\infty \frac{1}{1+\omega^2 R^2 C^2} \frac{d\omega}{2\pi} = \frac{W}{4RC}$$

We can write:

$$\boxed{\langle V_{out}^2(t) \rangle = \frac{W}{4RC}} = W \Delta f_n$$ → equivalent bandwidth for noise $\Delta f_n = \frac{1}{4RC}$

$$\int_0^\infty \frac{1}{1+\omega^2 R^2 C^2} \frac{d\omega}{2\pi} \quad [\omega RC = x]$$

$$\downarrow \int_0^\infty \frac{1}{1+x^2} \frac{dx}{2\pi RC} = \frac{1}{2\pi RC} \cdot \frac{\pi}{2} = \frac{1}{4RC}$$

Note that, for the signal, the same network has a bandwidth $\Delta f_s = \frac{1}{2\pi\tau} = \frac{1}{2\pi RC}$

The noise at the output is given by the noise power spectral density times the bandwidth for noise

We need to calculate $\boxed{W}$

<u>Thermodynamics argument</u>

The capacitor has a single degree of freedom → in Th. equilibrium has an average energy $\langle U \rangle = \frac{1}{2} KT$ For a capacitor $U = \frac{1}{2} C V^2 \Rightarrow \langle U \rangle = \frac{1}{2} C \langle V_{out}^2(t) \rangle$

$$\langle U \rangle = \frac{1}{2} KT = \frac{1}{2} C \langle V_{out}^2(t) \rangle \Rightarrow \boxed{\langle V_{out}^2(t) \rangle = \frac{KT}{C}}$$

Then we get $\frac{W}{4RC} = \frac{KT}{C}$ $\boxed{W = 4KTR}$ ← noise power spectral density

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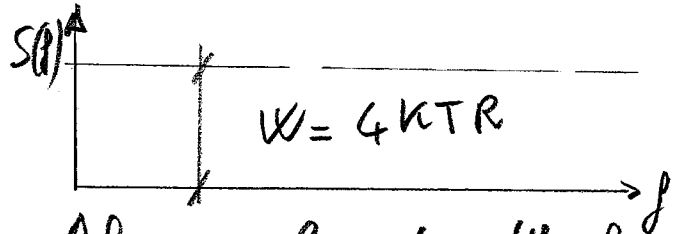
The Thermal (Johnson) noise is a White noise with Power spectral density $S(f) = 4KTR$



Every detector has a limited bandwidth Δf_N In general $\Delta f_N \to \Delta f_S$ ← bandwidth for the signal

<u>mean square value of the noise</u> ↔ <u>Average Power of the noise</u>

$$\langle V_J^2(t) \rangle = \langle P_N \rangle = 4KTR \cdot \Delta f$$ ← Bandwidth of the detector

<u>RMS noise</u>

$$V_J^{RMS} = \sqrt{\langle V_J^2(t) \rangle} = \sqrt{4KTR\,\Delta f}$$ ← <u>VOLTAGE</u>

$$I_J^{RMS} = \sqrt{\langle I_J^2(t) \rangle} = \frac{1}{R}\sqrt{\langle V_J^2(t) \rangle} = \sqrt{\frac{4KT\,\Delta f}{R}}$$ ← <u>CURRENT</u>

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